

Statement of Work

Support for NASA awards to develop and fly the EXoplanet Climate Infrared TElescope (EXCITE)

Summary:

The contractor shall provide engineering support for NASA award 18-APRA18-0075, entitled "Probing Exoplanet Atmospheric Physics Using The EXoplanet Climate Infrared TElescope (EXCITE)." The work will center on laboratory instrumentation, hardware integration, and operational support, including some or all of the following:

1. Design and implement custom laboratory instrumentation for testing and analyzing mercury cadmium telluride (HgCdTe) detector systems.
2. Provide high throughput, real-time analysis solutions and off-line data reduction procedures.
3. Integrate ancillary data sources and equipment and provide appropriate equipment synchronization solutions.
4. Provide testing, training and laboratory operational support.
5. Support an instrument deployment field campaign.
6. Document system design and maintain version control.

Required skills/knowledge; additional technical requirements

- Expert in readout and control of devices HgCdTe detectors, ACADIA controllers, MACIE detector electronics.
- Expert in developing systems for control of detector systems and cryogenic systems that operate from suborbital platforms, including stratospheric balloons and aircraft.
- Expert in reading out visible light camera systems that are used for pointing feedback for high stability infrared pointing platforms.
- Expert in deploying suborbital instrumentation from field settings, including NASA's Columbia Scientific Balloon Facility (NASA/CSBF) in Palestine, TX and Ft. Sumner, NM.

Period of Performance:

01/01/2026 - 12/31/2027